

FAHAD BIN SHAHABUDHEEN

Mechanical Engineer — Medical & IVD Instrumentation — Product Development

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Professional Summary

Mechanical Engineer with > 3 years of experience in mechanical product development, medical/IVD instrumentation, and electromechanical systems. Experienced in SolidWorks, GD&T, tolerance stack-up, FEA, DFM/DFX, dFMEA, pFMEA, design verification, rapid prototyping, and engineering documentation. Hands-on experience with fluid handling, microfluidics, precision dispensing, thermal systems, automated sample processing, and optical/chemiluminescence detection.

Core Skills

Mechanical Design: Product Development, 3D CAD, GD&T, Tolerance Stack-up, Engineering Drawings, BOMs, Material Selection, Mechanisms, DFM/DFX

Product Development: Product Lifecycle Management, Design Changes, Engineering Change Documentation, Design Reviews, Feasibility Studies, Verification & Validation

Risk & Verification: dFMEA, pFMEA, Risk Assessment, Risk Mitigation, Root Cause Analysis, Corrective Actions, Test Plans, Test Reports

Analysis: Structural/Thermal/Flow FEA, ANSYS Mechanical, ANSYS APDL, ANSYS Fluent, Thermodynamics, Heat Transfer

Instrumentation: IVD/Biochemical Instrumentation, Microfluidics, Fluid Handling, Pneumatic/Vacuum Systems, Precision Dispensing, Thermal Control, Electromechanical Systems, Automation

Manufacturing: CNC, EDM, Sheet Metal, Injection Moulding, Vacuum Casting, Thermoforming, FDM/SLA, Rapid Prototyping, Tooling & Fixtures

Quality / Regulatory: ISO 13485, ISO 14971, IEC 60601, IEC 61000-series EMI/EMC

Experience

Indian Institute of Science (IISc) / ShanMukha Innovations (SMI)

Oct. 2025 – Present

Senior Systems Engineer

Bengaluru, Karnataka

- Lead mechanical development of diagnostic, scientific, and IVD-related instrumentation spanning automated sample processing, fluid handling, optical/chemiluminescence detection, ultrafast PCR, and precision dispensing.
- Design mechanical and electromechanical assemblies using SolidWorks and Fusion 360, including GD&T, tolerance analysis, stack-ups, component selection, BOMs, and manufacturing drawings.
- Apply dFMEA and pFMEA to identify and mitigate product and process risks, including modification and qualification of IQC/FQC processes for medical-device hardware.
- Develop microfluidic and electromechanical assemblies for automated DNA extraction, fluorescence readout, sample washing, and precision dispensing; achieved positioning precision below $100\mu\text{m}$ and residual volumes below $0.5\mu\text{L}$.
- Perform structural and thermal FEA using ANSYS Mechanical, ANSYS APDL, and ANSYS Fluent, supported by first-principles calculations, to evaluate mechanical integrity, thermal behaviour, and dimensional stability.
- Design and validate precision fixtures and rigs for vibration testing, optical alignment, rotational mass-balance measurement, and positioning-error characterisation; developed machine-vision metrology achieving approximately $\pm 10\mu\text{m}$ measurement accuracy.
- Led mechanical and thermal architecture of a battery-operated ultrafast photothermal PCR system, optimising thermal mass, heat transfer, insulation, and mechanical interfaces for rapid thermal cycling.
- Designed, fabricated, characterised, and tested a SiPM-based low-light chemiluminescence detector integrating mechanical, optical, and thermal subsystems; achieved approximately 1000 counts/s LLOD with 10% 1σ CV and operation up to 5M counts/s.
- Performed EMI/EMC testing against applicable IEC 61000-series requirements and implemented mechanical/electrical design modifications to address identified compliance issues on an existing diagnostic instrument.
- Worked within ISO 13485 and ISO 14971 quality/risk-management frameworks, incorporating IEC 60601 requirements into medical-device design and verification.
- Conduct engineering investigations, root-cause analysis, risk assessments, design reviews, and corrective design changes for issues identified during fabrication, assembly, testing, and verification.

Cre-AID Labs

Dec. 2024 – Sept. 2025

Chief Technology Officer

Bengaluru, Karnataka

- Led multidisciplinary product development across mechanical, electrical, embedded, manufacturing, and software engineering for medical-device hardware.

- Conducted Pre-FEED and FEED analyses, established engineering standards and DFM frameworks, and coordinated manufacturing, procurement, engineering, and external fabrication activities.
- Coordinated design changes and engineering improvements across mechanical, electrical, manufacturing, and procurement teams, from technical assessment through implementation and verification.

Cre-AID Labs

Dec. 2023 – Nov. 2024

R&D Lead Engineer

Bengaluru, Karnataka

- Developed FEA workflows and material-characterisation methods for functional FDM components and established DFM methodologies across CNC, EDM, injection moulding, thermoforming, vacuum casting, and composite fabrication.
- Led development of medical and electromechanical products including negative pressure wound-therapy pumps, EMG monitors, heated moulding platforms, orthoses, dynamic splints, and exoskeleton systems.
- Developed electromechanical components including servomotor drives, BLDC motors, linear actuators, and precision micro-encoders.
- Developed engineering documentation and supported technical reviews, recruitment, and cross-functional product-development activities.

Cre-AID Labs

July 2023 – Nov. 2023

R&D Engineer

Bengaluru, Karnataka

- Designed, prototyped, and tested a 3D-printed elbow exoskeleton; reduced prototype expenditure by 70% and development time by 45% through CAE-driven optimisation.
- Designed and tested rehabilitation hardware, developed engineering-analysis software, and supported electronics and embedded development for electromechanical products.

Freelance

July 2021 – July 2023

Engineering Consultant

Kochi, Kerala

- Provided engineering feasibility studies, mechanical analysis, DFM feedback, and technical consulting for robotics, marine, and aerospace projects.

Technical Skills

CAD / Engineering: SolidWorks, PTC Creo, Fusion 360, Siemens NX, Engineering Drawings, GD&T, Tolerance Stack-up, BOMs

Simulation: ANSYS Mechanical, ANSYS APDL, ANSYS Fluent, OpenFOAM, CalculiX, Structural FEA, Thermal FEA, Flow Analysis

Manufacturing: CNC Machining, Turning, EDM, Precision Grinding, Sheet Metal, Injection Moulding, Vacuum Casting, Thermoforming, FDM, SLA

Electromechanical / Programming: Fluid Handling, Pneumatic/Vacuum Systems, Thermal Control, Microfluidics, Precision Dispensing, Motors, Actuators, Automation, Python, C/C++

Education

Kerala Technological University

Sep. 2017 – July 2021

Bachelor of Technology in Mechanical Engineering

Kochi, Kerala

Selected Achievements

- Published a conference paper in a peer-reviewed journal.
- National finalist, E-Yantra Ideas Competition, IIT Bombay.
- National finalist, Open Innovation Contest, VSSC, ISRO.
- First place, ASME Wind Power Challenge.